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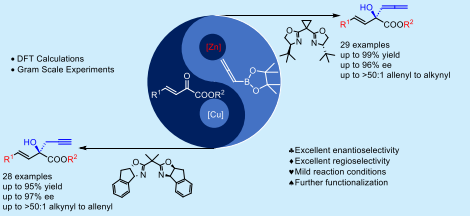
Enantio- and Regioselective Propargylation and Allenylation of β,γ-Unsaturated α-Ketoesters with Allenyl Boronate Regulated by Chiral Copper/Zinc

Shuai-Ru Cheng, Ni Zhao, Wei Ma, Guo-Rong Ma, and Pei Wang*

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ABSTRACT: A method for asymmetric propargylation and allenylation of β,γ-unsaturated α-ketoesters catalyzed by Box-Cu/Box-Zn using a commercial organoboron reagent is reported. The reaction was demonstrated to be broadly tolerant of functional groups, resulting in a series of propargyl and allenyl tertiary alcohols in excellent yields with high enantio- and regioselectivities. Furthermore, the proposed mechanism was supported by density functional theory calculations. Gram-scale experiments as well as the synthesis of various chiral tertiary alcohols and glycol derivatives demonstrated the practicality of this synthetic strategy.



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Homopropargyl and allenyl alcohols are common motifs in biologically active natural products and pharmaceutical compounds as well as essential synthetic units.[1] Although numerous methodologies for synthesizing these alcohols exist, the most straightforward and effective method remains the addition of propargyl or allenyl reagents to carbonyl compounds. Chiral secondary homopropargyl and allenyl alcohols have been successfully synthesized through the addition of various propargyl and allenyl metal species (Sn-, Zn-, B-, Si-, or In-based) to aldehydes, respectively.[2] However, the synthesis of chiral tertiary homopropargyl and allenyl alcohols has received less attention due to the inferior activity and selectivity of ketones compared to aldehydes. This study mainly focuses on highly active ketones, such as fluoro-substituted ketones,[3] isatins,[4] and methyl ketones.[5]

Allenylboronic acid pinacol ester is commercially available and relatively stable, which has been widely used for enantioselective propargylation and allenylation of ketones to synthesize homopropargyl tertiary alcohols in recent years.[3a,b,4a−d,5a,b] However, allenylmetals can undergo a metallotropic rearrangement resulting in either homopropargyl alcohol or allenyl alcohol when reacting with a carbonyl compound in a metal-catalyzed system,[6] as a metal−boron exchange reaction occurs between allenylboronic acid pinacol ester and the metal catalyst. Nevertheless, selectively synthesizing chiral homopropargyl or allenyl alcohols by controlling metallotropic rearrangement remains challenging.

The Fandrick group obtained racemic allenyl alcohols with high selectivity (93:7 to 99:1 allenyl:alkynyl) by zinc-catalyzed allenylations of aldehydes and ketones.[7] Subsequently, a practical and highly site-selective allenylation with propargyl boronates was developed. Site selectivity can be guided by BPE ligands and boronate substituents, with high selectivity (≤99:1) obtained with either propargyl or allenyl adducts (Scheme 1A).[8] However, the enantioselectivities of the propargyl and allenyl adducts were not studied. In 2015, the Kobayashi group synthesized racemic tertiary allenyl and homopropargyl alcohols with high selectivities from the same starting materials simply by altering the zinc reagent loading. In addition, a catalytic asymmetric variant of the allenylation of ketoesters was conducted, but only tertiary allenyl alcohol with 68% ee was obtained through allenylation of ketoesters (Scheme 1B).[9] Jarvo et al. developed silver-catalyzed addition reactions of allenylboronic acid pinacol esters to ketoesters to provide homopropargyl alcohols with excellent enantioselectivity, but accompanied by poor regioselectivity (2.5:1 homopropargyl:allenyl) (Scheme 1C).[10] Therefore, using readily available allenylboronic acid pinacol ester and ketoesters to highly selectively synthesize optically active tertiary homopropargyl and allenyl alcohols by controlling the metallotropic rearrangement remains a challenge.

Herein, we carried out the reaction between allenylboronic acid pinacol ester and β,γ-unsaturated α-ketoesters under mild conditions and obtained tertiary homopropargyl and allenyl alcohols with high enantioselectivity and regioselectivity by adjusting the metal reagent and solvent, respectively (Scheme 1D).

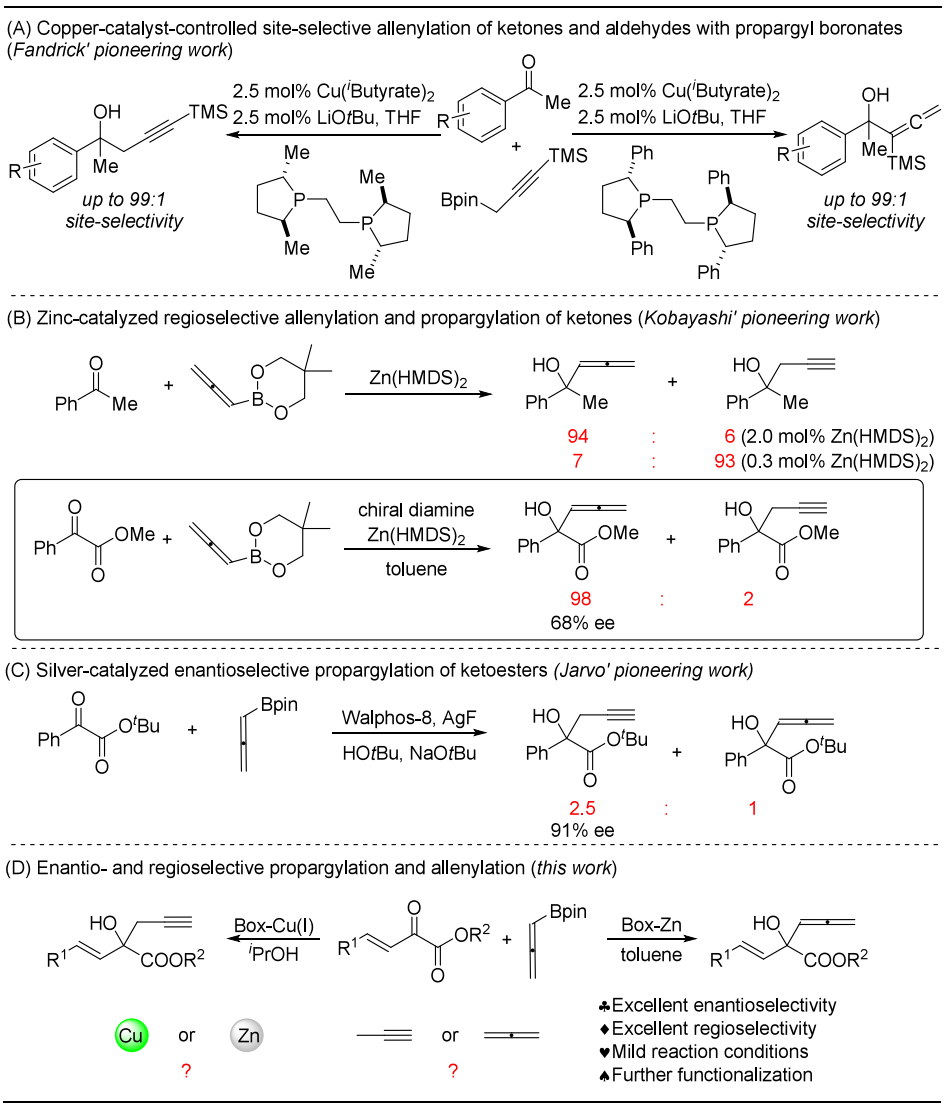
To assess the feasibility of the reaction, we initiated the identification of the reaction parameters using **2a** with **3** as prototype substrates with copper catalysts. In the presence of 12 mol % ligand and 10 mol % $Cu(MeCN)_4BF_4$ in isopropanol

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Scheme 1. Catalytic Propargylation and Allenylation of Ketoesters and Ketones

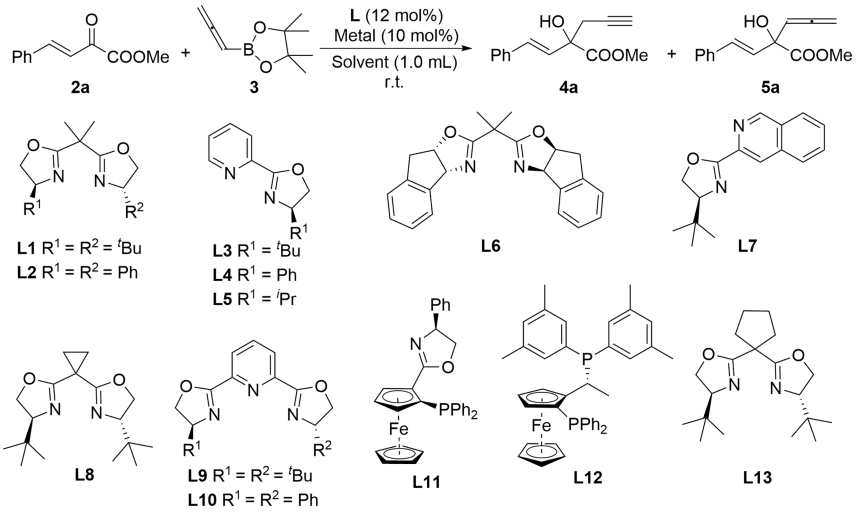


at room temperature, bisoxazoline ligand **L1** mainly afforded racemic homopropargyl alcohol with high yield (Table 1, entry 1). Ligand **L2** obtained a poor ee value (entry 2). When ligand **L3** was employed, a good yield and moderate enantioselectivity were achieved (entry 3). However, the large *tert*-butyl substituent on the oxazoline ring was replaced with phenyl, and the ee value decreased dramatically, although the yield remained high (entry 4). Ligand **L7** and ferrocene ligand **L11** provided unsatisfactory results (entries 6 and 8, respectively). Subsequently, we again focused on other bisoxazoline ligands (entries 5 and 7). **L6** gave the most favorable outcomes, affording product **4a** in 66% yield, with high enantioselectivity and good selectivity for propargylation over allenylation. Then, we evaluated the effect of copper salts, solvent, and temperature on the reaction system (as detailed in the Supporting Information). According to the literature,[5b,10] NaO^tBu can effectively promote the reaction. Nevertheless, NaO^tBu was ineffective for this reaction, and the selectivity decreased significantly (as detailed in the Supporting Information). Surprisingly, allenyl alcohol was obtained as the major product when zinc salt was used. Subsequently, we optimized the conditions for the zinc-catalyzed allenylation, as detailed in Table 1 and the Supporting Information.

With the optimal reaction conditions in hand, various β,γ-unsaturated α-ketoesters were investigated to study the application scope of the developed methodology (Table 2). In these examples, all reactions afforded chiral homopropargyl tertiary alcohols (**4a**−**4q**) with high enantioselectivity and good regioselectivity for propargylation over allenylation. A range of functional groups and substitution patterns were tolerated. The substitution of R^2 by ethyl and isopropyl groups with large steric hindrance generated homopropargyl products (**4b** and **4c**) with correspondingly good yields, high enantioselectivities, and excellent regioselectivities, suggesting that ester groups with large groups were also well tolerated in this reaction system. Except for **4g**, **4o**, and **4q**, homopropargyl alcohols **4d**−**4p** had good yields, which might be due to the generation of byproducts caused by the alcohol transesterification of unsaturated ketoesters and isopropanol. Then, we expanded the tolerance of alkynyl ketoesters with various substituents in this reaction system. The results indicated that **4aa**−**4ah** exhibited high enantioselectivity and excellent regioselectivity. Notably, although moderate yields

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Table 1. Optimization of the Reaction Conditions[a]



entry	metal salts, L	time (h)	4a:5a[b]	yield (%)[c]	ee (%)
1	$Cu(MeCN)_4BF_4$, **L1**	20	12:1	99	<1
2	$Cu(MeCN)_4BF_4$, **L2**	144	>20:1	59	20
3	$Cu(MeCN)_4BF_4$, **L3**	20	11:1	97	68
4	$Cu(MeCN)_4BF_4$, **L4**	35	8:1	99	6
5	$Cu(MeCN)_4BF_4$, **L6**	48	>20:1	66	95
6	$Cu(MeCN)_4BF_4$, **L7**	10	14:1	97	56
7	$Cu(MeCN)_4BF_4$, **L8**	72	15:1	50	7
8	$Cu(MeCN)_4BF_4$, **L11**	72	5:1	63	3
9	$Zn(CF_3SO_3)_2$, **L1**	12	1:6	92	86
10	$Zn(CF_3SO_3)_2$, **L2**	120	1:5	47	22
11	$Zn(CF_3SO_3)_2$, **L5**	9	1:5	63	<1
12	$Zn(CF_3SO_3)_2$, **L6**	120	1:7	52	52
13	$Zn(CF_3SO_3)_2$, **L8**	0.5	1:6	93	93
14	$Zn(CF_3SO_3)_2$, **L9**	9	1:7	66	10
15	$Zn(CF_3SO_3)_2$, **L10**	48	1:16	69	<1
16	$Zn(CF_3SO_3)_2$, **L11**	9	1:24	70	10
17	$Zn(CF_3SO_3)_2$, **L12**	120	1:9	64	<1
18	$Zn(CF_3SO_3)_2$, **L13**	0.5	1:6	91	86

[a]Unless noted otherwise, reactions were performed with **2a** (0.25 mmol), **3** (0.5 mmol), **L** (0.03 mmol, 12 mol %), and metal salts (0.025 mmol, 10 mol %) in isopropanol (1.0 mL) under an argon atmosphere at rt. [b]For reactions of **4a** and **5a** catalyzed by copper salts and zinc salts, the **4a**:**5a** ratios were measured by chiral HPLC analysis after column chromatography purification. [c]The resulting yield was the isolated yield.

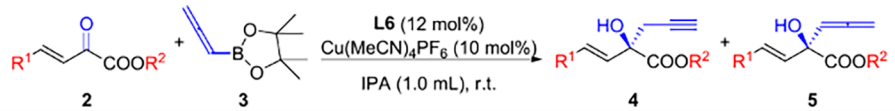
were observed in the case of *para*-electron-withdrawing substituents (**4ad**−**4af**), *β*-naphthyl (**4ag**) and *p*-biphenyl groups (**4ah**), they all achieved high enantioselectivity and good regioselectivity. Here, we have also applied copper-catalyzed homopropylation to the multistep synthesis of aliphatic product **4ai** from the corresponding ethynyl estradiol, with good yields, high enantioselectivities, and excellent regioselectivities. Aryl-substituted glyoxylate fails to produce the corresponding homopropargyl alcohol, whereas the alkyl-substituted glyoxylate gave **4aj** with high enantioselectivity and excellent regioselectivity. In the process, the absolute configuration of **4l** was determined to be *R* by X-ray.

We subsequently expanded the substrates for the asymmetric synthesis of tertiary allenyl alcohols with zinc catalysts (Table 3). A wide range of substrates bearing strong steric hindrance (**5b** and **5c**), both electron-donating groups (**5d**−**5g**, **5ab**, and **5ac**), halogenated (**5h**−**5l** and **5ad**−**5af**), heteroaryl (**5o** and **5p**), *β*-naphthyl (**5q** and **5ag**), styryl (**5r**), 4-biphenyl (**5s** and **5ah**), and aliphatic groups (**5ai**), can participate in the reaction. The desired tertiary allenyl alcohols were constructed in 78−99% yields with 66−96% ee. Good regioselectivity was obtained for all products, except **5o**, **5r**, and **5ai**. Aryl-substituted glyoxylate cannot produce allenyl alcohol, while alkyl-substituted glyoxylate can afford **5aj** with high enantioselectivity and good regioselectivity. **4l** was reduced to give highly enantioselective **4ak**, and **5l** was reduced by the same method; its reduction product was determined by HPLC to be highly enantioselective, and the times of the major and minor peaks are consistent with **4ak**. Therefore, the configuration of **5l** was determined to be *R*.

To elucidate the reaction mechanism, possible transition-state models were proposed. For Cu-catalyzed homopropargylation, (*R*)-**Cu-IM4** passes through (*R*)-**Cu-TS2**, crossing an energy barrier of 1.5 kcal mol^{-1} to generate the *R*-configuration product. For the blue pathway, **2a** was added to form complex (*R*)-**Cu-IM3** with **Cu-IM1**, and (*R*)-**Cu-IM3** passes through (*R*)-**Cu-TS1**, crossing an energy barrier of 3.2 kcal mol^{-1} to generate the *R*-configuration product (Scheme 2a). Compared to the blue pathway, the highest energy barrier for the green pathway is lower, indicating that green pathway is more favorable in terms of kinetics. For Zn-catalyzed allenylation, an energy barrier of 9.2 kcal mol^{-1} was crossed to form the *R*-

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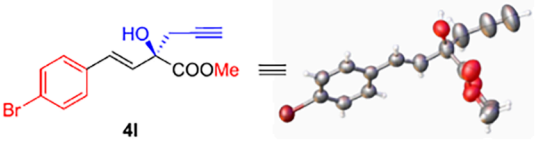
Table 2. Asymmetric Synthesis of Homopropargyl Alcohols[a–d]

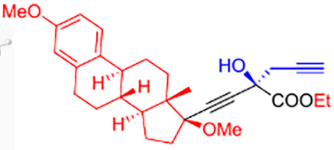


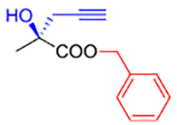
4a, R^1 = C_6H_5, R^2 = Me, 83% (95% ee), 43:1
4b, R^1 = C_6H_5, R^2 = Et, 82% (94% ee), 33:1
4c, R^1 = C_6H_5, R^2 = ^{i}Pr, 87% (92% ee), >50:1
4d, R^1 = 4-Me-C_6H_4, R^2 = Me, 69% (96% ee), 50:1
4e, R^1 = 2-OMe-C_6H_4, R^2 = Et, 80% (96% ee), 50:1
4f, R^1 = 3-OMe-C_6H_4, R^2 = Me, 83% (96% ee), 43:1
4g, R^1 = 4-OMe-C_6H_4, R^2 = Me, 50% (94% ee), 13:1

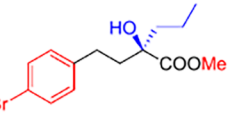
4h, R^1 = 4-F, 3-Me-C_6H_3, R^2 = Me, 78% (95% ee), >50:1
4i, R^1 = 4-F-C_6H_4, R^2 = Me, 84% (97% ee), 50:1
4j, R^1 = 2-Cl-C_6H_4, R^2 = Me, 68% (96% ee), >50:1
4k, R^1 = 3-Cl-C_6H_4, R^2 = Me, 65% (95% ee), >50:1
4l, R^1 = 4-Br-C_6H_4, R^2 = Me, 91% (95% ee), 50:1
4m, R^1 = 4-CF_3-C_6H_4, R^2 = Me, 90% (95% ee), 50:1
4n, R^1 = 2-thienyl, R^2 = Me, 81% (95% ee), >50:1
4o, R^1 = 1-naphthyl, R^2 = Me, 51% (96% ee), >50:1
4p, R^1 = 2-naphthyl, R^2 = Me, 70% (96% ee), 50:1
4q, R^1 = C_6H_5CH=CH, R^2 = Me, 52% (95% ee), 30:1

4aa, R^1 = C_6H_5, 71% (94% ee), >50:1
4ab, R^1 = 4-Me-C_6H_4, 81% (95% ee), >50:1
4ac, R^1 = 4-OMe-C_6H_4, 75% (96% ee), >50:1
4ad, R^1 = 4-F-C_6H_4, 49% (93% ee), 50:1
4ae, R^1 = 4-Cl-C_6H_4, 56% (94% ee), 50:1
4af, R^1 = 4-Br-C_6H_4, 44% (94% ee), 50:1
4ag, R^1 = 2-naphthyl, 37% (89% ee), >50:1
4ah, R^1 = *p*-biphenyl, 35% (92% ee), 33:1


4l

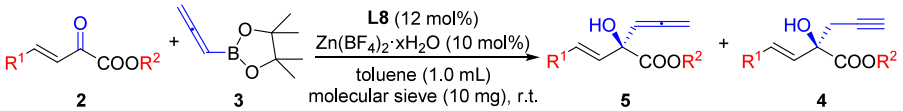

4ai 95% (92% ee) >50:1


4aj 70% (94% ee) =50:1


4ak[e] 100%[f] (95% ee)

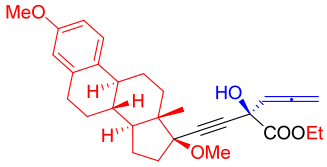
[a]Unless noted otherwise, the resulting homopropargyl alcohols were produced under the corresponding optimal conditions. [b]The resulting yield was the isolated yield. [c]The ratios were measured according to the ^{1}H NMR spectrum after column chromatography purification. [d]The enantioselectivity of homopropargyl alcohols was measured by chiral HPLC analysis after column chromatography purification. [e]**4ak** was obtained from **4l** by reduction of Pd/C and hydrogen. [f]Conversion is reported, which was directly measured by HPLC and ^{1}H NMR without column chromatography separation.

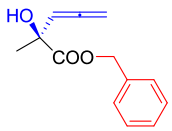
Table 3. Asymmetric Synthesis of Allenyl Alcohols[a–d]



5a, R^1 = C_6H_5, R^2 = Me, 95% (95% ee), 21:1
5b, R^1 = C_6H_5, R^2 = Et, 95% (95% ee), 25:1
5c, R^1 = C_6H_5, R^2 = ^{i}Pr, 81% (95% ee), 25:1
5d, R^1 = 4-Me-C_6H_4, R^2 = Me, 93% (95% ee), 25:1
5e, R^1 = 2-OMe-C_6H_4, R^2 = Et, 79% (92% ee), >50:1
5f, R^1 = 3-OMe-C_6H_4, R^2 = Me, 88% (96% ee), 21:1
5g, R^1 = 4-OMe-C_6H_4, R^2 = Me, 94% (95% ee), 33:1
5h, R^1 = 4-F, 3-Me-C_6H_3, R^2 = Me, 93% (96% ee), 21:1

5i, R^1 = 4-F-C_6H_4, R^2 = Me, 84% (94% ee), 23:1
5j, R^1 = 2-Cl-C_6H_4, R^2 = Me, 91% (93% ee), 21:1
5k, R^1 = 3-Cl-C_6H_4, R^2 = Me, 82% (96% ee), 17:1
5l, R^1 = 4-Br-C_6H_4, R^2 = Me, 78% (95% ee), 14:1
5m, R^1 = 4-CF_3-C_6H_4, R^2 = Me, 91% (96% ee), 21:1
5n, R^1 = 4-NO_2-C_6H_4, R^2 = Me, 83% (96% ee), 23:1
5o, R^1 = 2-furyl, R^2 = Me, 78% (95% ee), 4:1
5p, R^1 = 2-thienyl, R^2 = Me, 95% (95% ee), 17:1
5q, R^1 = 2-naphthyl, R^2 = Me, 83% (96% ee), 23:1
5r, R^1 = C_6H_5CH=CH, R^2 = Me, 91% (95% ee), 4:1
5s, R^1 = *p*-biphenyl, R^2 = Me, 93% (96% ee), 21:1


5ai 91% (84% ee) =10:1


5aj 75% (93% ee) =17:1

5aa, R^1 = C_6H_5, 91% (71% ee), 50:1
5ab, R^1 = 4-Me-C_6H_4, 98% (70% ee), 25:1
5ac, R^1 = 4-OMe-C_6H_4, 98% (73% ee), 33:1

5ad, R^1 = 4-F-C_6H_4, 98% (69% ee), 33:1
5ae, R^1 = 4-Cl-C_6H_4, 90% (72% ee), 25:1
5af, R^1 = 4-Br-C_6H_4, 91% (71% ee), 33:1
5ag, R^1 = 2-naphthyl, 99% (66% ee), 50:1
5ah, R^1 = *p*-biphenyl, 94% (72% ee), 33:1

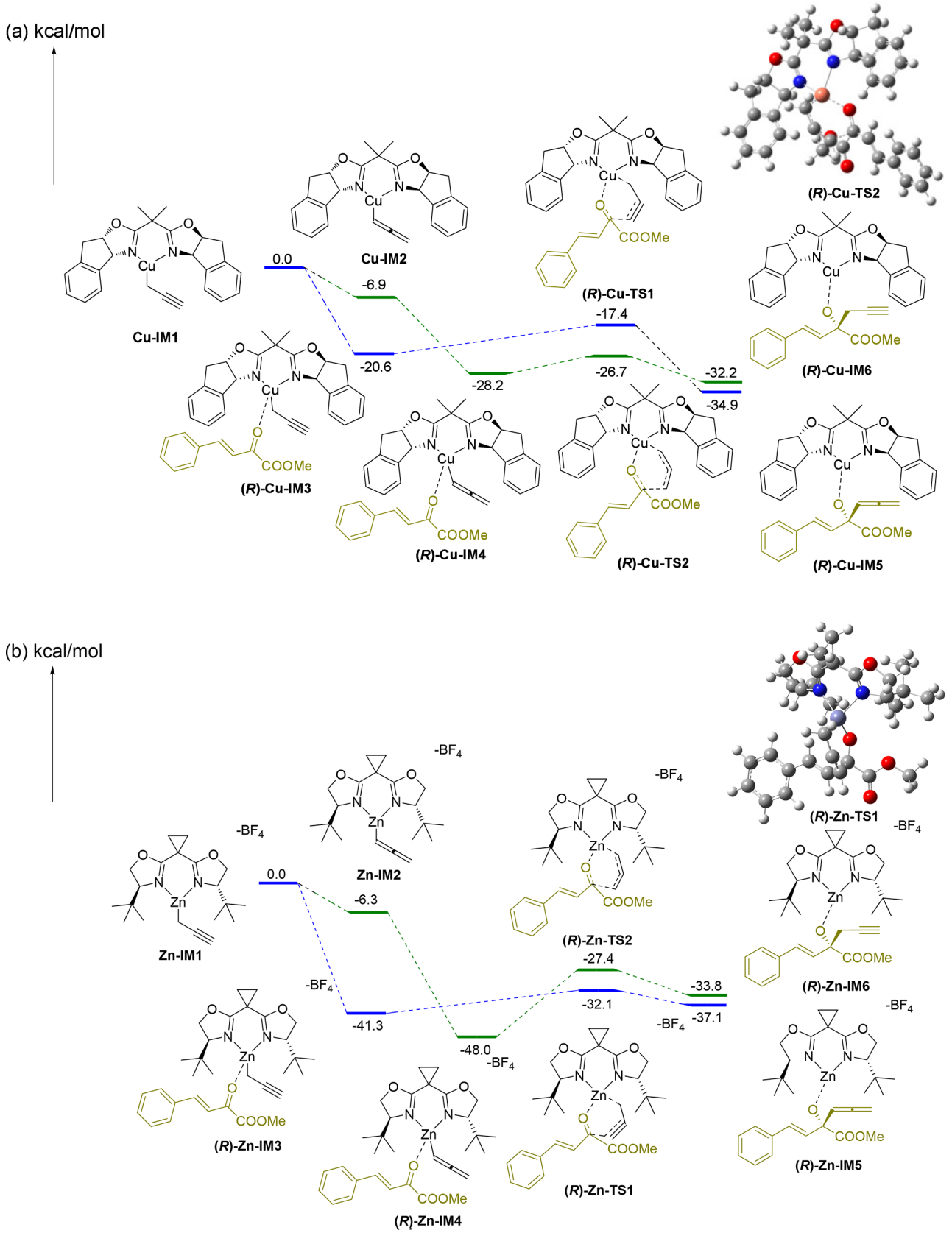
[a]Unless noted otherwise, the resulting allenyl alcohols were produced under the corresponding optimal conditions. [b]The resulting yield was the isolated yield. [c]The ratios were measured according to the ^{1}H NMR spectrum after column chromatography purification. [d]The enantioselectivity of allenyl alcohols was measured by chiral HPLC analysis after column chromatography purification.

configuration allenyl alcohol [(*R*)-**Zn-IM3** to (*R*)-**Zn-TS1**], and (*R*)-**Zn-IM4** crosses the energy barrier of 20.6 kcal mol^{-1} through (*R*)-**Zn-TS2** to form (*R*)-**Zn-IM6**. Compared with that of the green pathway, the maximum energy barrier of the blue pathway is lower, so the blue pathway is better kinetically. Because the isomerization between **Cu-IM1** and **Cu-IM2** is present (see Figure S1), propargyl alcohols are obtained, and the transformation adheres to the Curtin–Hammett principle. In contrast, **Zn-IM1** is hard to isomerize to **Zn-IM2** (see Figure S2), so **Zn-IM1** reacts to form the allenyl alcohols. The three-dimensional structures of transition states (*R*)-**Cu-TS2** and (*R*)-**Zn-TS1** are also demonstrated in Scheme 2.

To further demonstrate the potential of this homopropargylation and allenylation reactions, Cu-catalyzed homopropargylation and Zn-catalyzed allenylation were both conducted on a gram scale, leading to homopropargyl alcohol **4a** (86% yield,

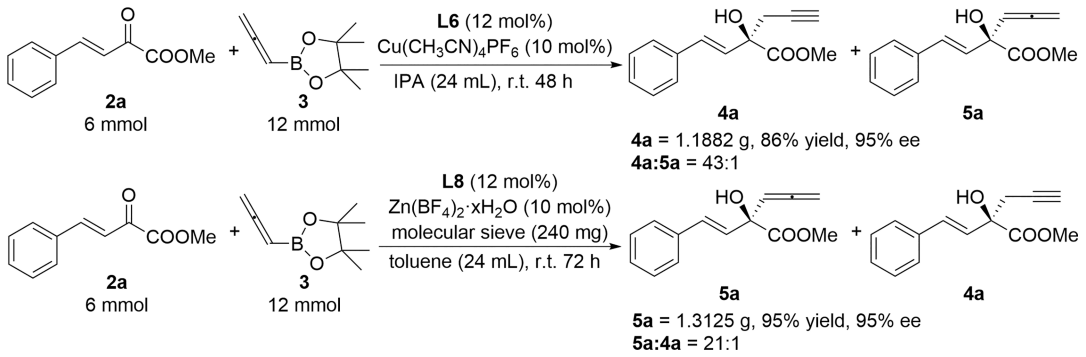
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Scheme 2. Gibbs Free Energy Profiles for the Copper- and Zinc-Catalyzed Enantioselective Propargylation and Allenylation of 2a with 3



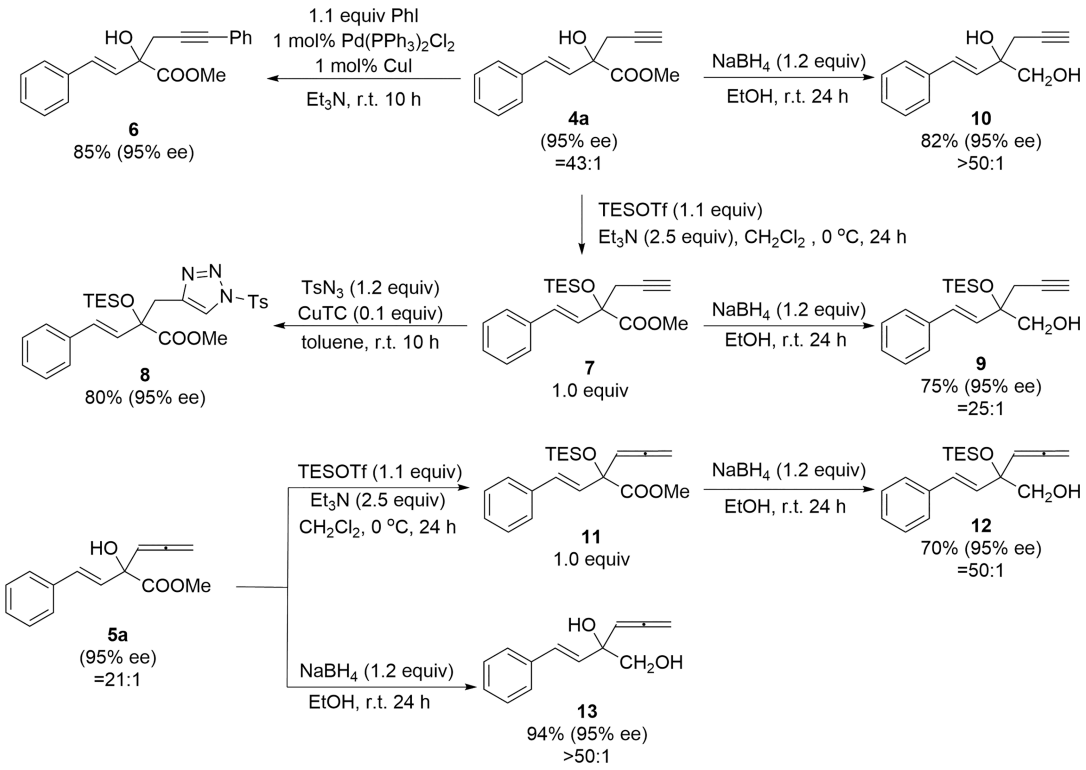
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Scheme 3. Gram-Scale Reactions[a,b]



[a]The ratios were measured according to the ^{1}H NMR spectrum after column chromatography purification. [b]The enantioselectivity was measured by chiral HPLC analysis after column chromatography purification.

Scheme 4. Synthetic Applications[a,b]



[a]The ratios were measured according to the ^{1}H NMR spectrum after column chromatography purification. [b]The enantioselectivity was measured by chiral HPLC analysis after column chromatography purification.

95% ee, 43:1 **4a**:**5a**) and allenyl alcohol **5a** (95% yield, 95% ee, 1:21 **4a**:**5a**) without erasing the yield, enantioselectivity, or regioselectivity (Scheme 3).

Homopropargyl and allenyl alcohols are significant synthons that can be converted into various chiral tertiary alcohols and glycol derivatives (Scheme 4). Sonogashira coupling of **4a** in the presence of 1.0 mol % $PdCl_2(PPh_3)_2$ and 1.0 mol % CuI affords coupling product **6**, enabling introduction of highly diversified substituent at the alkyne. **4a** with TESOTf generated **7**, and then click reaction promoted by CuTc provided 1,2,3-triazole **8**. **7** was reduced by $NaBH_4$ to produce alcohol **9**. Transformation of **4a** through $NaBH_4$ led to the ester compounds being reduced to glycol **10**. The reaction of allenyl alcohol **5a** with TESOTf in ethanol furnished **11**, and then its reduction by $NaBH_4$ produced **12**. **5a** was directly reduced to **13** by $NaBH_4$.

In summary, an asymmetric propargylation and allenylation of β,γ-unsaturated α-ketoesters with a commercial organoboron reagent was developed under Box-Cu/Box-Zn catalysis. A series of unsaturated allenylation and allenyl tertiary alcohols were afforded in excellent yields with high enantio- and regioselectivities. The proposed mechanism was supported by a density functional theory (DFT) calculation. The synthetic utility of this strategy was demonstrated by gram-scale experiments and the synthesis of various chiral tertiary alcohol and glycol derivatives.

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■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c03845.

FAIR data, including the primary NMR FID files, for compounds **2a−2t**, **2aa−2aj**, and **2ai-1** (ZIP)
FAIR data, including the primary NMR FID files, for compounds **4a−4i** (ZIP)
FAIR data, including the primary NMR FID files, for compounds **4j−4q** (ZIP)
FAIR data, including the primary NMR FID files, for compounds **4aa−4ak** (ZIP)
FAIR data, including the primary NMR FID files, for compounds **5a−5i** (ZIP)
FAIR data, including the primary NMR FID files, for compounds **5j−5s** (ZIP)
FAIR data, including the primary NMR FID files, for compounds **5aa−5aj** (ZIP)
FAIR data, including the primary NMR FID files, for compounds **6**, **8**, **9**, **10**, **12**, and **13** (ZIP)
Procedures, spectroscopic data, DFT computational details, and crystallographic data for all compounds (PDF)

Accession Codes

Deposition Number 2361745 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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